

Problem: 01

Problem Definition: Given $F(A,B,C) = A'BC + AB'C + ABC' + ABC$, a sum-of-products (SOP) expression in three variables, simplify it to its minimal form and implement it with logic gates.

Objective: To reduce a 4-term SOP expression to its simplest 3-term form using the consensus/redundancy technique (splitting the ABC term to pair with each minterm) and to confirm the reduction via K-map grouping.

Simplification:

$$F(A, B, C) = A'BC + AB'C + ABC' + ABC$$

$$= A'BC + ABC + AB'C + ABC + ABC' + ABC$$

$$= BC + AC + AB$$

$$= AB + AC + BC$$

Karnaugh Map:

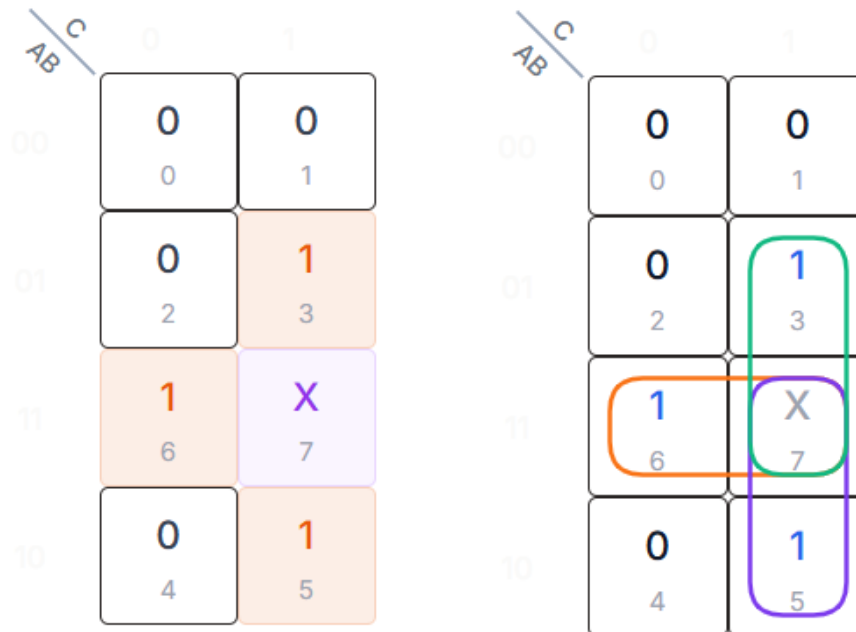


Figure 1: Karnaugh Map

Circuit Diagram:

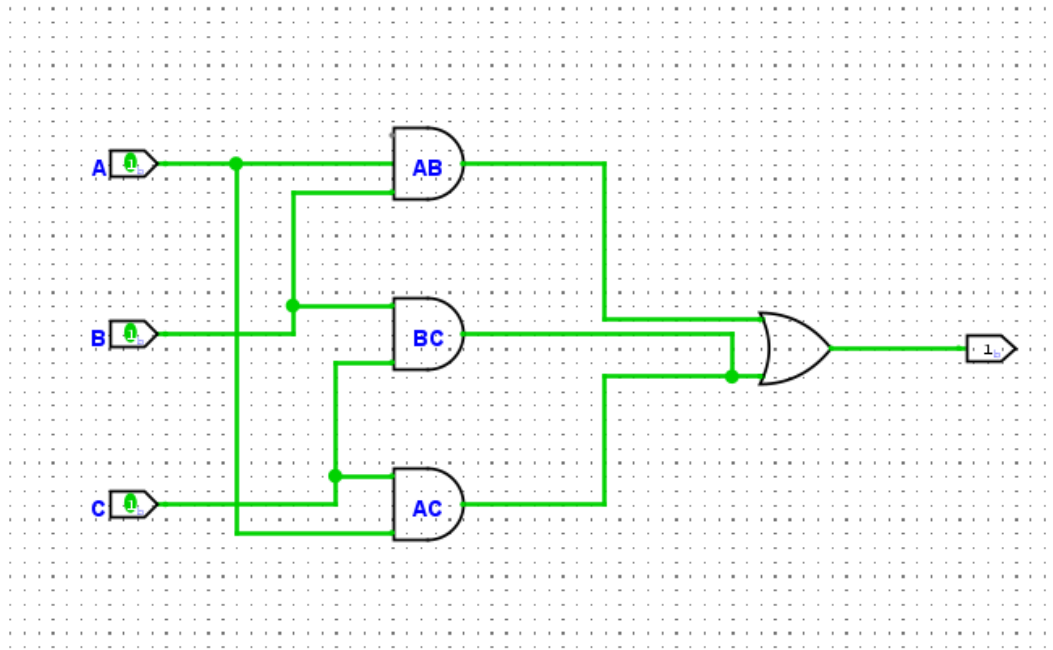


Figure 2: $AB+BC+AC$

Discussion: $F1$ has four minterms: $A'BC$, $AB'C$, ABC' , ABC . Since $ABC = ABC + ABC + ABC$ (idempotent law), the term can be duplicated and paired with each of the other three terms to eliminate one variable per pair: $A'BC+ABC = BC$, $AB'C+ABC = AC$, $ABC'+ABC = AB$. This gives $F1 = AB+AC+BC$. On the K-map, these four minterms occupy adjacent cells that form three overlapping pairs of 2, each pair eliminating one variable — this is the classic case where a single minterm (ABC , at cell 111) is "reused" in multiple groups, which is allowed and often necessary for a minimal cover. The final circuit needs only 3 AND gates (2-input each) and one 3-input OR gate, versus 4 AND gates (3-input) and a 4-input OR gate for the unsimplified form.

Problem: 02

Problem Definition: Given $F(A,B,C) = A(A+B)(A+B+C)$, a product-of-sums-like expression, simplify it using absorption laws and implement the result.

Objective: To apply the absorption law ($X + XY = X$, and its generalization $X(X+Y)=X$) repeatedly to show that the entire expression collapses to a single literal.

Simplification:

$$\begin{aligned} 2. F(A, B, C) &= A(A+B)(A+B+C) \\ &= (A + AB)(A+B+C) \\ &= A + AB + AC + AB + AB + ABC \\ &= A + AB + AC + ABC \\ &= A + AB + AC \\ &= A(1 + B + C) \\ &= A \end{aligned}$$

Karnaugh Map:

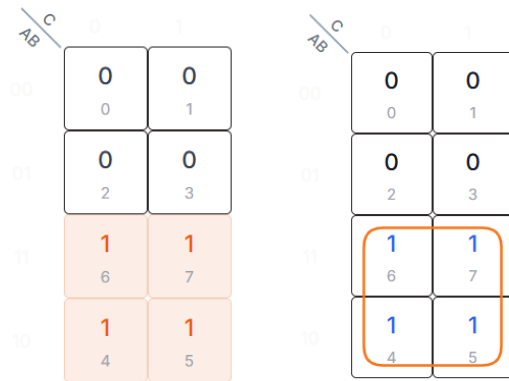


Figure 3: Karnaugh Map

Circuit Diagram:



Figure 4: A

Discussion: The absorption law states $A(A+B) = A$. Applying it first to $A(A+B)$ leaves A , and then $A(A+B+C)$ again reduces to A by the same law. So $F2 = A$ regardless of B and C . This means on the K-map, $F2 = 1$ for every cell where $A = 1$ (a group of 4 covering the entire $A=1$ half of the map), independent of B and C . The circuit is trivial: no gates are required at all — the output is simply wired to input A . This example demonstrates how algebraic simplification can eliminate an entire circuit's worth of gates when redundant conditions are present.

Problem: 03

Problem Definition: Given $F(A,B,C) = (A+(BC))'(AB+ABC)$, simplify the expression using De Morgan's theorem and Boolean postulates, and show that it reduces to the constant 0.

Objective: To apply De Morgan's law to a complemented sum term and show algebraically that ANDing complementary terms $(A \cdot A')$ produces a null function, then confirm this with an all-zero K-map.

Simplification:

$$3. F(A, B, C) = (A+(BC))'(AB + ABC)$$

$$= AB(A + (BC))'$$

$$= AB(A + B' + C')$$

$$= AB (A'BC)$$

$$= 0$$

Karnaugh Map:

		C	
		0	1
AB	00	0 0	0 1
	01	0 2	0 3
	11	0 6	0 7
	10	0 4	0 5

Figure 5: Karnaugh Map

Circuit Diagram:

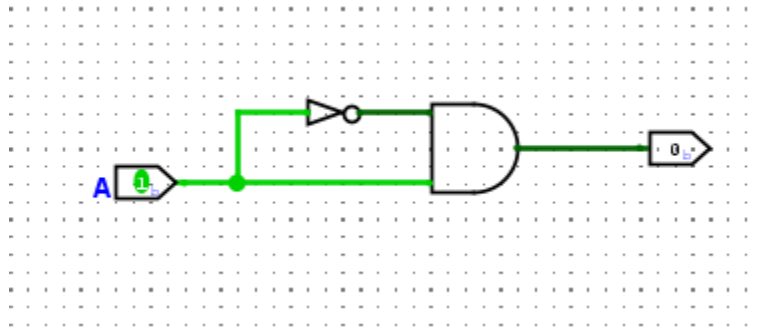


Figure 6: 0

Discussion: First, $AB+ABC$ simplifies to AB by absorption ($AB(1+C)=AB$). Next, De Morgan's theorem is applied twice to $(A+(BC))'$: since $(BC)' = B'+C'$, we get $A+B'+C'$, and negating that sum gives $A' \cdot B \cdot C$. Multiplying the two simplified factors: $F3 = AB \cdot A'BC = (AA')BBC = 0 \cdot BC = 0$, since a variable ANDed with its complement is always 0. On the K-map, this means every cell is 0 — there is no minterm coverage at all. In the circuit, this is shown as the output permanently tied to logic LOW; no combination of A, B, C can produce a 1. This problem illustrates how algebraic manipulation can reveal that a seemingly complex expression is actually a contradiction (always-false function).

Problem: 04

Problem Definition: Given $F(A,B,C) = (B'(A+B) + (A+B)(A+B'))B'$, simplify using distribution and complement laws, and implement the resulting 2-literal expression.

Objective: To expand nested product terms, apply the complement law ($X \cdot X' = 0$) and identity law ($X \cdot 1 = X$) to reduce a 3-variable-looking expression down to a 2-variable product term, then verify the reduction is independent of C.

Simplification:

$$\begin{aligned} 4. F(A, B, C) &= (B'(A+B) + (A+B)(A+B'))B' \\ &= (AB' + A + AB' + AB)B' \\ &= (A + AB + AB')B' \\ &= AB'(1 + \dots) \\ &= AB' \end{aligned}$$

Karnaugh Map:

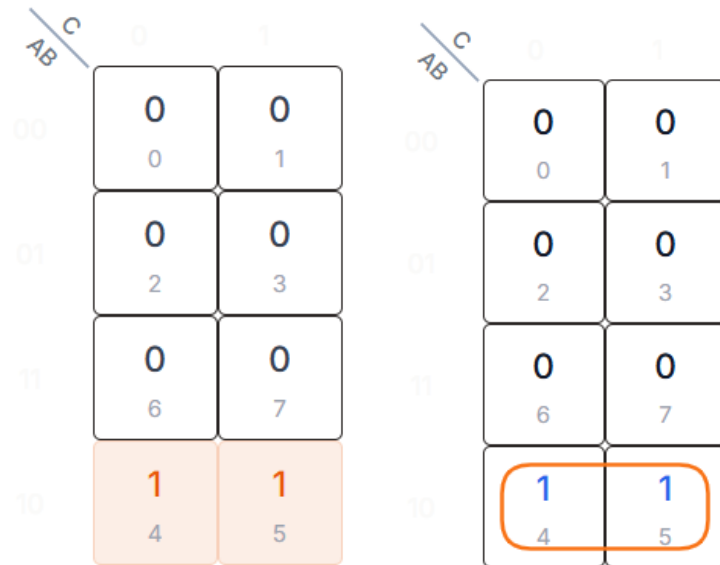


Figure 7: Karnaugh Map

Circuit Diagram:

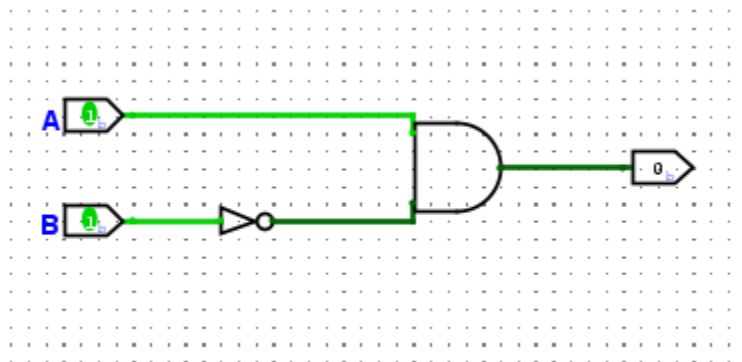


Figure 8: AB'

Discussion: Expanding $B'(A+B)$ gives $AB'+BB' = AB'$ (since $BB'=0$). Expanding $(A+B)(A+B')$ gives $A+AB'+AB+BB' = A(1+B'+B) = A$. Adding these: $AB'+A = A(1+B') = A$. Multiplying by the outer B' : $F4 = A \cdot B' = AB'$. Notably, C never appears in the original expression, so the function is automatically independent of C — on the K-map this shows up as two full columns (for $C=0$ and $C=1$) both marked wherever $A=1$, $B=0$, forming a group of 2 that spans the C dimension. The final circuit needs only one NOT gate (to get B') and one AND gate, making it the simplest of the four circuits.

